

Task B1 – Python Environment Setup

As part of the project setup phase, a requirements.txt file was created to standardize the Python environment used by all team members. This file specifies the exact versions of the main libraries required for data processing, visualization, and machine learning. By using the same dependency versions, the team can avoid compatibility issues and ensure that code behaves consistently across different computers.

The selected libraries are:

- Pandas: data manipulation and analysis.
- NumPy: numerical computing and array operations.
- Scikit-Learn: machine learning algorithms and model evaluation.
- Matplotlib: data visualization and plotting.
- Seaborn: statistical data visualization built on top of Matplotlib.
- OpenPyXL: reading and writing Excel files, which is essential since the project uses historical IoT data provided as Excel extracts.
- Jupyter: interactive notebooks for data exploration and experimentation.
- Joblib: saving and loading trained machine learning models efficiently.

To use the environment, team members simply execute the following command in their Python environment :

```
pip install -r requirements.txt
```

This automatically installs all required libraries with the specified versions, ensuring a reproducible and shared development environment throughout the project.